

Robust forecast evaluation under extreme shocks

Fabrizio Iacone^{1,2} and Andrea Viselli³

¹Università degli Studi di Milano ²University of York ³BI Norwegian Business School

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Website: andreaviselli.it

Email: andrea.viselli@bi.no

Extreme observations over brief time periods can reduce the power of the Diebold–Mariano (DM) test. When the instability window is known in advance, we propose the winsorised DM* test to recover the usual DM test’s power.

1. Introduction

Large, short-lived shocks (the “instability”) in forecast loss differentials can inflate the long-run variance estimate and mask the predictive advantage of the better forecasting method. As a result, the ordinary DM test may fail to detect systematic differences in predictive accuracy.

In this study, we propose a simple correction: winsorise the loss differential over the known instability period. Unlike removing these observations, winsorisation preserves information about both their magnitude and direction.

2. The winsorised statistic

For a generic loss differential

$$d_t = L(e_{1,t}) - L(e_{2,t}), \quad t_{DM} = \sqrt{T} \bar{d} / \hat{\sigma},$$

We consider the data generating process

$$d_t = \delta_1 T^{-1/2} + \delta_2 T^\alpha I_t(\tau) + u_t,$$

where $I_t(\tau) = 1$ only at $t = \lfloor \tau T \rfloor$, with τ known in advance.

For $\delta_2 \neq 0$, shocks with $\alpha < 1/2$ do not affect the limiting performance of the DM test. Extreme shocks, instead, affect both its numerator and denominator and drive power towards zero for reasonable theoretical size (Iacone et al., 2025).

To cap large shocks, define

$$d_{\inf} = \inf_{t \neq \lfloor \tau T \rfloor} \{d_t\}, \quad d_{\sup} = \sup_{t \neq \lfloor \tau T \rfloor} \{d_t\},$$

and winsorise the instability observation as

$$d_{\lfloor \tau T \rfloor}^* = \begin{cases} d_{\inf}, & d_{\lfloor \tau T \rfloor} < d_{\inf}, \\ d_{\sup}, & d_{\lfloor \tau T \rfloor} > d_{\sup}, \end{cases}$$

while $d_{\lfloor \tau T \rfloor}^* = d_{\lfloor \tau T \rfloor}$ otherwise. The statistic computed from $\{d_1, \dots, d_{\lfloor \tau T \rfloor}^*, \dots, d_T\}$ is denoted by t_{DM}^* and the test by DM*.

2.1. Asymptotic properties

The long-run variance is estimated using the Bartlett kernel.

Assumptions (Giacomini and White, 2006).

(i) u_t is mixing with ϕ of size $-r/(2r-2)$, $r \geq 2$; or α of size $-r/(r-2)$, $r > 2$.

(ii) $E(|u_t|^{2(r+\zeta)}) < \infty$, for $\zeta > 0$ and all t .

(iii) $\text{Var}(T^{-1/2} \sum_{t=1}^T u_t) > 0$ for all T sufficiently large.

In addition, we assume that $\sigma^2 = \lim_{T \rightarrow \infty} \text{Var}(T^{-1/2} \sum_{s=1}^T u_s)$ exists.

Theorem 1. Under these conditions,

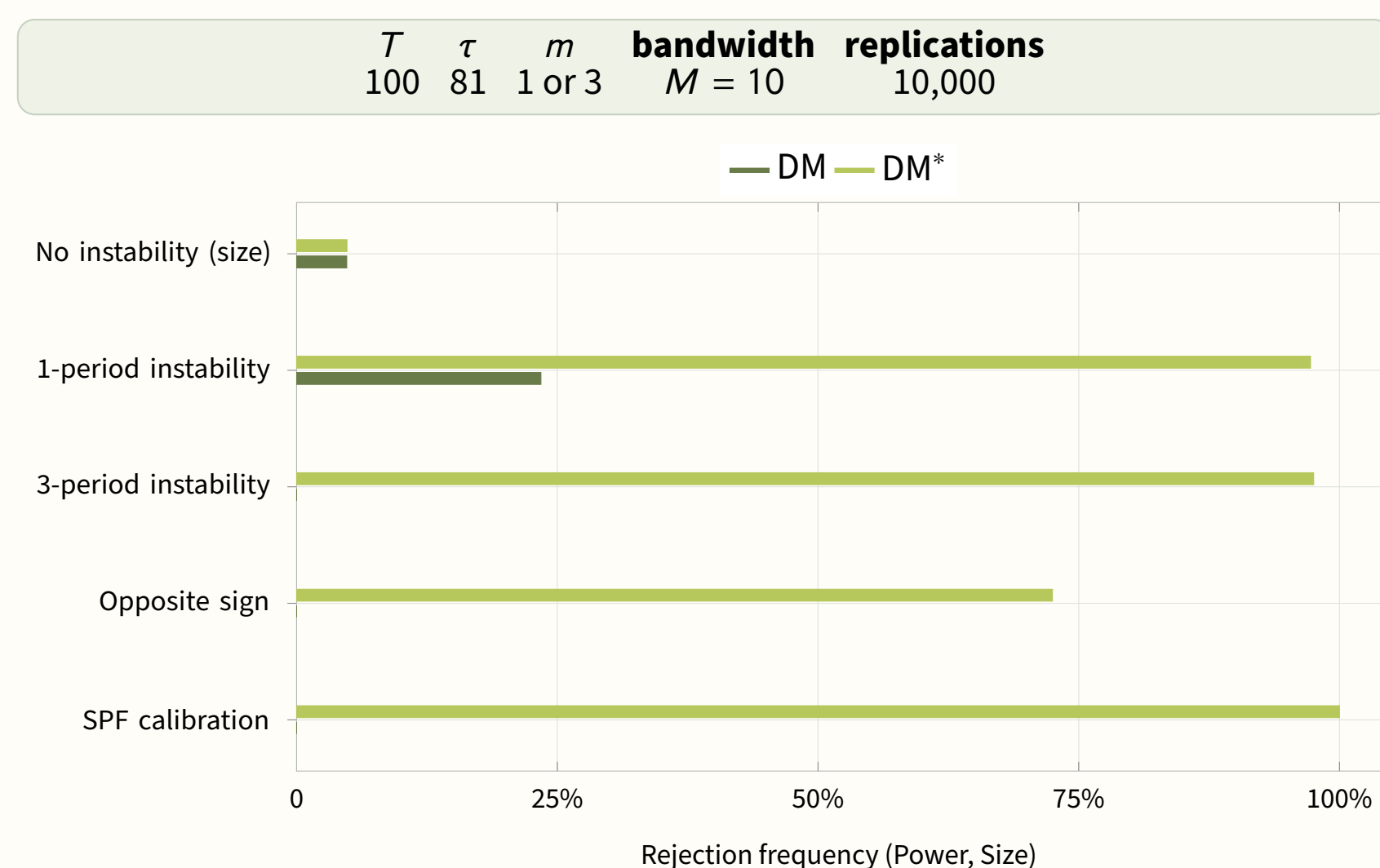
$$t_{DM}^* \xrightarrow{d} Z + \frac{\delta_1}{\sigma},$$

and we recover the local power of the DM statistic.

3. Monte Carlo study

We study the winsorised DM test under

$$d_t = \kappa + \theta I_t(\tau; m) + u_t, \quad u_t = \phi u_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, 1).$$



We report only **selected designs**. The “no instability” case corresponds to the empirical size. For the one- and three-period instability, we use $\kappa = -0.4$ and $\theta = -40$. In the case with opposite sign, we use $\kappa = 0.4$, $\theta = -40$, and $m = 3$.

Design calibrated on Covid-19

$\kappa = -0.8$, $\theta = -80$, $\phi = 0.25$, $m = 3$.

This simulation mimics the setup of the empirical application.

DM

0%

DM*

100%

4. Empirical application to US real GDP growth

SPF nowcasts are compared with a zero-growth benchmark over 2000:Q1–2024:Q4. The Covid-19 instability is 2020:Q1–2020:Q3.

MSE: SPF

4.51

MSE: zero-growth

31.17

DM

1.458

DM*

6.147

The 5% critical value is 2.261 (Coroneo and Iacone, 2020).

5. Conclusions

In applied work, DM* should be reported alongside the standard DM statistic as a transparent robustness check when the sample contains rare, severe, and well-identified shocks.

Diebold and Mariano (1995); Giacomini and White (2006); Coroneo and Iacone (2020); Iacone, Rossini and Viselli (2026).

DM Test Dashboard

Scan to run the Monte Carlo simulation with different parameter choices and compare DM with the winsorised DM* statistic.

andreaviselli.it/dm-dashboard/

